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# SPECIFICATION FOR APPROVAL

DATE：

CUSTOMER：

PART NAME： Thick Film Automotive Chip Resistors

CUSTOMER'S DWG. NO.：

CUSTOMER'S PART NO.：

PDC PART NO.： **FWF SERIES APPROVED**

DESCRIPTION.：

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| ACTION / RESULT | " V " | CUSTOMER'S SIGNATURE | NOTE |
|---|---|---|---|
| FULL APPROVED | | | |
| CONDITIONAL APPROVED | | | |
| REJECTED | | | |

| OUR ACTION | SIGNATURE |
|---|---|
| PREPARED By | Jenny Tseng |
| CHECKED By | Tony Chou |
| APPROVED By | Byron Tsai |

| CUSTOMER SIGNATURE FOR ACCEPTANCE |
|---|
| |

PDC

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# FWF series. (AEC-Q200)
# Automotive Grade General Purpose Thick-film Lead Free Chip Resistors

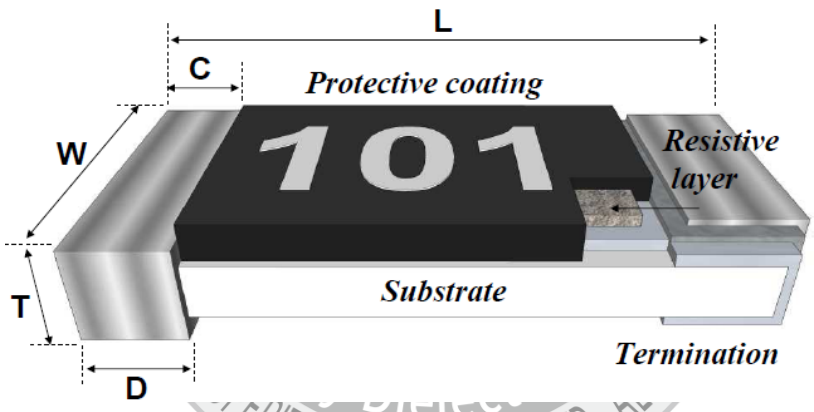
## 1. Features

- Meet AEC-Q200 test for Automotive industry.
- Suitable for lead free soldering.
- Compatible with wave and reflow soldering.
- Anti-sulfur products and 100% CCD inspection.
- RoHS compliant &. Halogen free.

## 2. Applications

- Automotive industry.
- Digital meter, Consumer electronics, M/B.
- Portable electronics devices.

## 3.Dimension and Construction



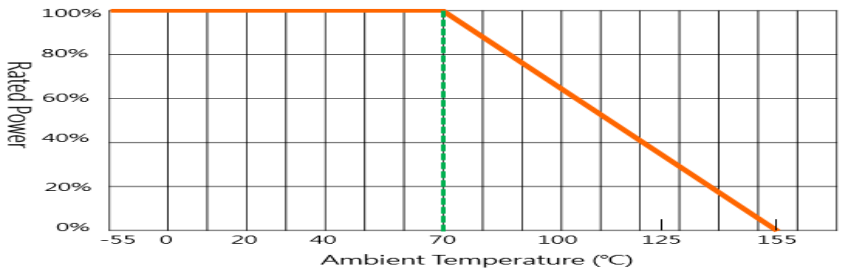

Unit：mm

| Type | L | W | C | D | T |
|---|---|---|---|---|---|
| FWF02 | 1.00±0.05 | 0.50±0.05 | 0.20±0.10 | 0.25±0.10 | 0.35±0.05 |
| FWF03 | 1.60±0.10 | 0.80±0.10 | 0.30±0.20 | 0.30±0.20 | 0.45±0.10 |
| FWF05 | 2.00±0.10 | 1.25±0.10 | 0.40±0.20 | 0.40±0.20 | 0.50±0.10 |
| FWF06 | 3.10±0.10 | 1.60±0.10 | 0.50±0.25 | 0.50±0.25 | 0.55±0.10 |
| FWF12 | 3.10±0.10 | 2.60±0.10 | 0.50±0.25 | 0.50±0.25 | 0.55±0.10 |
| FWF20 | 5.00±0.20 | 2.50±0.20 | 0.60±0.25 | 0.60±0.25 | 0.60±0.10 |
| FWF25 | 6.40±0.20 | 3.20±0.20 | 0.60±0.25 | 0.90±0.25 | 0.60±0.15 |

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## 4.Power Derating Curve

Operating Temperature Range: -55 to +155℃




## 5.Rating

5.1 General Resistance

| Type<br>**FWF** | Size | Power Rating at 70℃ | Max. RCWV | Max. Overload Voltage | Resistance Tolerance | Temperature Coefficient (ppm/℃) | Resistance Range Min. | Resistance Range Max. | Standard Resistance Values |
|---|---|---|---|---|---|---|---|---|---|
| **02**<br>**02-W** | 0402 | 1/16W<br>1/10W | 50V | 100V | ±1%(F)<br>±5%(J) | ±200 | >1MΩ | 10MΩ | E-96(F)<br>E-24(J) |
| | | | | | | ±100 | >10Ω | 1MΩ | |
| | | | | | | -200~+400 | 0，1Ω | 10Ω | |
| **03**<br>**03-W** | 0603 | 1/10W<br>1/10W | 75V | 150V | ±1%(F)<br>±5%(J) | ±200 | >1MΩ | 10MΩ | E-96(F)<br>E-24(J) |
| | | | | | | ±100 | >10Ω | 1MΩ | |
| | | | | | | -200~+400 | 0，1Ω | 10Ω | |
| **05**<br>**05-W** | 0805 | 1/8W<br>1/4W | 150V | 300V | ±1%(F)<br>±5%(J) | ±200 | >1MΩ | 10MΩ | E-96(F)<br>E-24(J) |
| | | | | | | ±100 | >10Ω | 1MΩ | |
| | | | | | | -200~+400 | 0，1Ω | 10Ω | |
| **06**<br>**06-W** | 1206 | 1/4W<br>1/4W | 200V | 400V | ±1%(F)<br>±5%(J) | ±200 | >1MΩ | 10MΩ | E-96(F)<br>E-24(J) |
| | | | | | | ±100 | >10Ω | 1MΩ | |
| | | | | | | -200~+400 | 0，1Ω | 10Ω | |
| **12**<br>**12-W** | 1210 | 1/2W<br>1/2W | 200V | 400V | ±1%(F)<br>±5%(J) | ±200 | >1MΩ | 10MΩ | E-96(F)<br>E-24(J) |
| | | | | | | ±100 | >10Ω | 1MΩ | |
| | | | | | | -200~+400 | 0，1Ω | 10Ω | |

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# FWF series. (AEC-Q200)
# Automotive Grade General Purpose Thick-film Lead Free Chip Resistors

| Type<br>**FWF** | Size | Power Rating at 70℃ | Max. RCWV | Max. Overload Voltage | Resistance Tolerance | Temperature Coefficient ( ppm/℃) | Resistance Range | | Standard Resistance Values |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min. | Max. | |
| **20** | 2010 | 3/4W | 200V | 400V | ±1%(F) | ±200 | >1MΩ | 10MΩ | E-96(F) |
| **20-W** | | 3/4W | | | ±5%(J) | ±100 | >10Ω | 1MΩ | E-24(J) |
| | | | | | | ±200 | 0，1Ω | 10Ω | |
| **25** | 2512 | 1W | 250V | 500V | ±1%(F) | ±200 | >1MΩ | 10MΩ | E-96(F) |
| **25-W** | | 1W | | | ±5%(J) | ±100 | >10Ω | 1MΩ | E-24(J) |
| | | | | | | ±200 | 0，1Ω | 10Ω | |

5.2 0 Ohm, Jumper

| Type<br>**FWF** | Size | Description | Max. Rated Current | Resistance |
|---|---|---|---|---|
| 02 | 0402 | Zero Ohm，Jumper | ≦ 1A | ＜50mΩ |
| 03 | 0603 | Zero Ohm，Jumper | ≦ 1A | ＜50mΩ |
| 05 | 0805 | Zero Ohm，Jumper | ≦ 2A | ＜50mΩ |
| 06 | 1206 | Zero Ohm，Jumper | ≦ 2A | ＜50mΩ |
| 12 | 1210 | Zero Ohm，Jumper | ≦ 3A | ＜50mΩ |
| 20 | 2010 | Zero Ohm，Jumper | ≦ 3A | ＜50mΩ |
| 25 | 2512 | Zero Ohm，Jumper | ≦ 3A | ＜50mΩ |

Notes:

1. RCWV is Rated Voltage, V=$\sqrt{P*R}$ or Max. Working Voltage whichever is lower.
2. V：Working Voltage(V)，P：Rated Power (W)，R：Resistance Value(Ω)
3. Please consider keep the surface temperature do not exceed 105℃ when working.

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## 6.Part Number

| Type | Size | Tolerance | Packing | Watt | R Value (GM) | TCR | Special Code |
|---|---|---|---|---|---|---|---|
| **FWF** | **02** :0402<br>**03** :0603<br>**05** :0805<br>**06** :1206<br>**12** :1210<br>**20** :2010<br>**25** :2512 | **F** :±1%<br>**J** :±5% | **Paper Tape：**<br>**0402.0603.**<br>**0805.1206**<br>**T** : 5Kpcs<br>**V** : 10Kpcs<br>**W** : 20Kpcs<br>**Plastic Tape：**<br>**2010.2512**<br>**P** : 4Kpcs<br>**X** : 8Kpcs<br>**Y** : 16Kpcs | **-**:<br>As<br>Rating<br>Info | **XXX**<br>**XXXX**<br>**5%:**<br>3 digits<br>**1%:**<br>4 digits | **-**:<br>As<br>Rating<br>Info | Null means<br>Standard<br>**W** :<br>Anti-sulfur<br>H2S<br>1000ppm |

Example：

**FWF05FT-1004**

➔0805 size, tolerance 1%, paper tape, 1/8W, 1 MΩ.

**FWF03JT-102**

➔0603 size, tolerance 5%, paper tape, 1/10W, 1 KΩ.

**FWF03FT-1004-W**

➔0603 size, tolerance 1%, paper tape, 1/10W, 1 MΩ, Special Anti-Sulfur.

## 7.Marking/Soldering

**Resistance value identify：**

E24 ±5%：3 Digits marking to identify the resistance value

0603/0805/1206/2010/2512




301 → 30 x $10^1$ = 300Ω

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E24/E96 ±1%：4 Digits marking to identify the resistance value

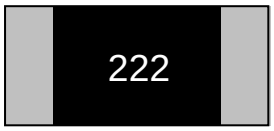
0805/1206/2010/2512




1542 → 154 x $10^2$ = 15.4 KΩ

E24 ±1%：3 Digits marking to identify the resistance value

0603




222 → 22 x $10^2$ = 2.2 KΩ

E96 ±1%：3 Digits marking to identify the resistance value

0603

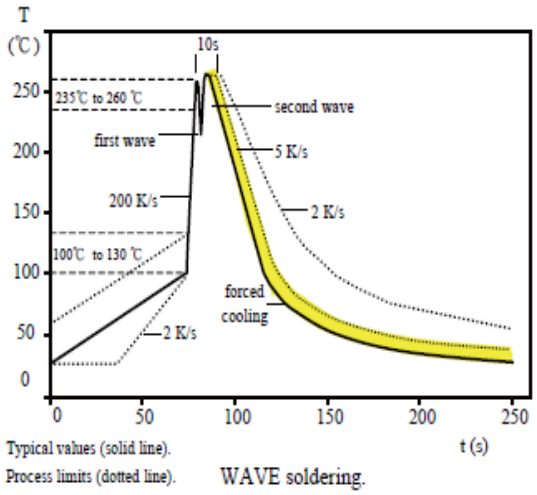



01B → Refer 0603 marking table = 1 KΩ

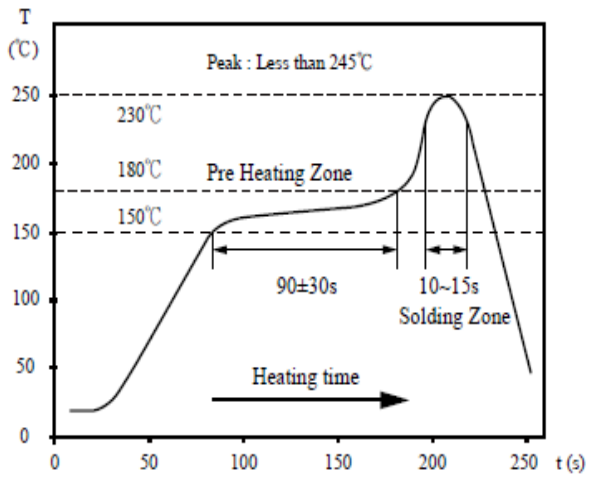
No marking of 0402 product.

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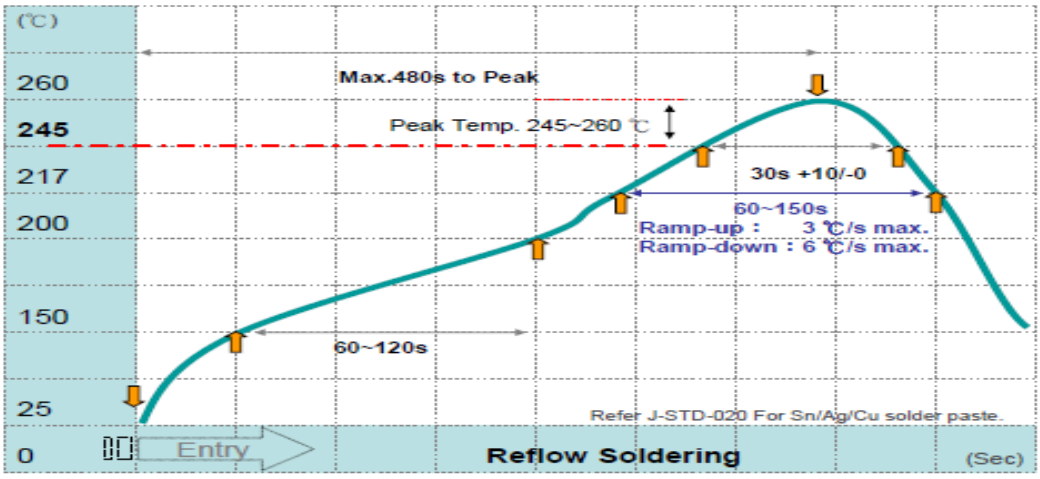
**Soldering Reference：Compatible for most industrial soldering request.**




WAVE soldering.

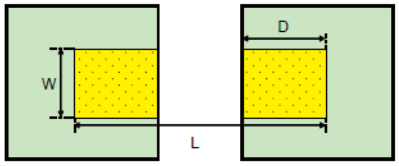



IR Reflow soldering.




**Recommend Solder Pad Dimensions :**

| Type | W | D | L |
|---|---|---|---|
| **FWF02** | 0.60 | 0.5 | 1.50 |
| **FWF03** | 0.90 | 1.00 | 3.00 |
| **FWF05** | 1.30 | 1.15 | 3.50 |
| **FWF06** | 1.80 | 1.30 | 4.70 |
| **FWF12** | 3.00 | 1.30 | 4.70 |
| **FWF20** | 3.00 | 1.50 | 6.80 |
| **FWF25** | 3.70 | 1.60 | 7.60 |




Unit:mm

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## 8. Reliability Performance (AEC-Q200)

| Test Item | Specification | Test Method (AEC-Q200. IEC 60115) |
|---|---|---|
| **DC Resistance** | F：±1%；J：±5% | **AEC-Q200 TABLE 7.1**<br>**IEC 60115-1 / JIS C 5201-1 , Clause 4.5** Measure the resistance Value. |
| **High Temperature Exposure (Storage)** | J：△R≦ ±(3%+0.1Ω)<br>F：△R≦ ±(1%+0.05Ω) | **AEC-Q200 TABLE 7.3**<br>1000 hrs. @ T=125℃. Unpowered.<br>Measurement at 24 ±2 hours after test conclusion. |
| **Temperature Cycling** | J：ΔR≦ ±(1%＋0.1Ω)<br>F：ΔR≦ ±(0.5%＋0.05Ω)<br>No mechanical damage. | **AEC-Q200 TABLE 7.4**<br>1000 Cycles (-55°C to +125°C). Measurement at 24±4 hours after test conclusion. |
| **Moisture Resistance** | J：△R≦ ±(1%+0.1Ω)<br>F：△R≦ ±(0.5%+0.05Ω) | **AEC-Q200 TABLE 7.6**<br>Test 65℃/80~100%RH/10Cycles.<br>Measurement at 24±2 hours after test conclusion.<br>(t=24hrs/cycle). |
| **Biased Humidity** | J：△R≦ ±(3%+0.1Ω)<br>F：△R≦ ±(1%+0.05Ω) | **AEC-Q200 TABLE 7.7**<br>1000 hours 85℃/85%RH.<br>10% of operating power.<br>Measurement at 24 ±2 hours after test conclusion. |
| **Operational Life** | J：△R≦ ±(3%+0.1Ω)<br>F：△R≦ ±(1%+0.05Ω) | **AEC-Q200 TABLE 7.8**<br>Test 1000hr @ TA=125℃ at specified rated power.<br>Measurement at 24±2 hours after test conclusion. |
| **External Visual** | No visual damage and refer PDC marking code. | **AEC-Q200 TABLE 7.9**<br>Inspect device construction, marking and workmanship. |
| **Physical Dimension** | Within the spec. | **AEC-Q200 TABLE 7.10**<br>Verify physical dimensions to the applicable device detail specification. |
| **Mechanical Shock** | Within product specification tolerance and no visible damage. | **AEC-Q200 TABLE 7.13**<br>Test Peak value:100g's,Wave:Hail-sine,<br>Duration:6ms,Velocity:12.3ft/sec. |

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| | | |
|---|---|---|
| **Vibration** | No mechanical damage. | **AEC-Q200 TABLE 7.14**<br>5 g's for 20 min., 12 cycles each of 3 orientations.<br>Test from 10-2000 Hz. |
| **Resistance to Solder Heat** | J：ΔR≦ ±(1%＋0.1Ω)<br>F：ΔR≦ ±(0.5%＋0.05Ω)<br>No mechanical damage. | **AEC-Q200 TABLE 7.15**<br>Solder dipping @ 270℃±5℃ for 10sec.±1sec. |
| **Thermal Shock** | J：ΔR≦ ±(1%＋0.1Ω)<br>F：ΔR≦ ±(0.5%＋0.05Ω)<br>No mechanical damage. | **AEC-Q200 TABLE 7.16**<br>-55 to 155℃/ dwell time 15min/ Max transfer time 20sec/ 300cycles. |
| **ESD** | ΔR≦ ±(1%＋0.1Ω)<br>No mechanical damage. | **AEC-Q200-002**<br>Test contact min. 0.5KV. |
| **Solder Ability** | Over 95% of termination must be covered with solder. | **AEC-Q200 TABLE 7.18**<br>a)Baking 155℃ 4H, dipping 235℃ 5s<br>b)Steam 1H, dipping 215℃ 5s<br>c)Steam 1H, dipping 260℃ 7s |
| **Flammability** | Refer UL-94. | **AEC-Q200 TABLE 7.20**<br>UL-94 V-0 or V-1 are acceptable |
| **Board Flex** | J：ΔR≦ ±(1%＋0.1Ω)<br>F：ΔR≦ ±(0.5%＋0.05Ω)<br>No mechanical damage. | **AEC-Q200 TABLE 7.21**<br>Bending 2mm 2512.2010.1210.1206,<br>3mm 0805.0603.0402. |
| **Terminal Strength** | No mechanical damage | **AEC-Q200 TABLE 7.22**<br>Force 1 Kg for 60 seconds. |
| **Short Time Overload** | J：ΔR≦ ±(1%＋0.1Ω)<br>F：ΔR≦ ±(0.5%＋0.05Ω) | **IEC 60115-1, Clause 4.13**<br>2.5 × Rated voltage or Max. overload voltage for 5 seconds |
| **Load Life Humidity** | J：△R≦ ±(3%+0.1Ω)<br>F：△R≦ ±(1%+0.05Ω) | **IEC 60115-1, Clause 4.24**<br>40±2℃ with relative humidity<br>90% ~ 95% D.C. rated voltage for<br>1.5 hours ON 30 minutes OFF.<br>Cycle repeated 1000 hours. |

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| | | |
|---|---|---|
| **Temperature Coefficient of Resistance (TCR)** | Within the spec. | **IEC 60115-1, Clause 4.8**<br>$T_1$ $T_2$<br>Test temperature： 25℃~ -55℃<br>25℃~ +155℃<br>TCR(ppm/℃) =<br>$(R_2-R_1)/R_1 \times 1 / (T_2-T_1) \times 10^6$ |
| **Load Life** | J：△R≦ ±(3%+0.1Ω)<br>F：△R≦ ±(1%+0.05Ω) | **IEC 60115-1, Clause 4.25**<br>Rated voltage for 1.5 hours for followed by a pause 0.5 hour at 70±2℃.<br>Cycle repeated 1000 hours. |
| **Insulation Resistance** | Between termination and coating must over 1000MΩ | **IEC 60115-1, Clause 4.6**<br>Test voltage：100±15V |
| **Anti-sulfur products require** | △R ≦ ± (1% + 0.1Ω)<br>No mechanical damage | **IEC 60068-2-43/42**<br>H2S：10-15 ppm，SO2：25±5 ppm，25±2℃ RH 75%<br>240Hrs<br>**Specialty**：-W，H2S：1000ppm 720Hrs |

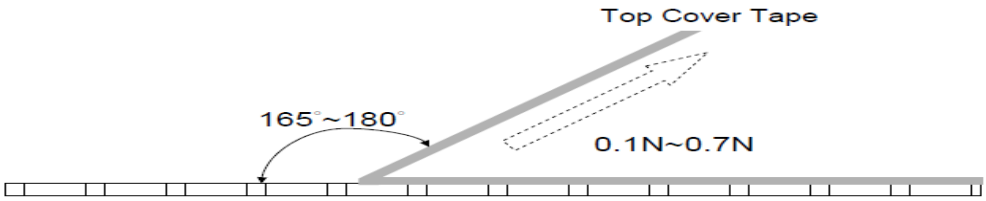
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## 9. PACKAGING

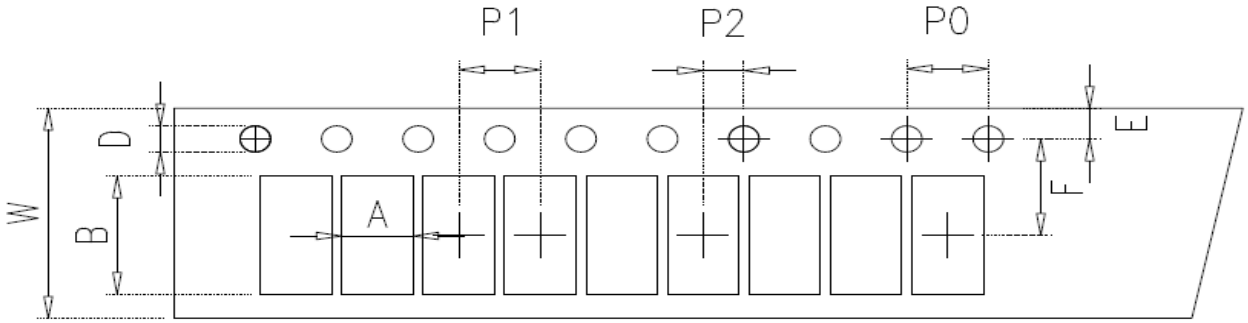
### 9.1 Peel Strength of Top Cover Tape

The peel speed shall be about 300 mm/min

The peel force of top cover tape shall between 0.1 to 0.7N




### 9.2 Tape Packaging Dimensions



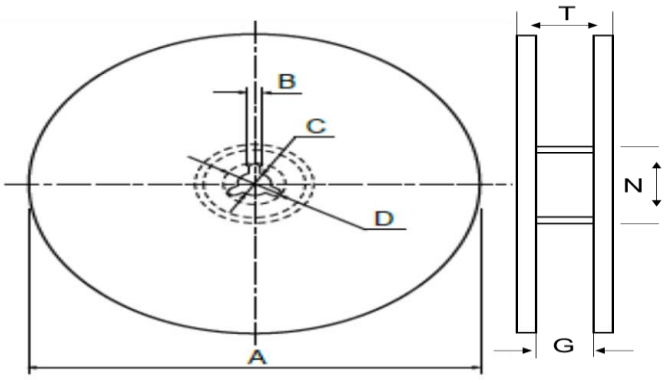

unit:mm

| Size | A | B | W | F | E | P1 | P2 | P0 | D |
|---|---|---|---|---|---|---|---|---|---|
| 0402 | 0.70±0.10 | 1.20±0.10 | 8.00±0.30 | 3.50±0.05 | 1.75±0.10 | 2.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |
| 0603 | 1.10±0.20 | 1.90±0.20 | 8.00±0.30 | 3.50±0.05 | 1.75±0.10 | 4.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |
| 0805 | 1.65±0.20 | 2.40±0.20 | 8.00±0.30 | 3.50±0.05 | 1.75±0.10 | 4.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |
| 1206 | 2.00±0.20 | 3.60±0.20 | 8.00±0.30 | 3.50±0.05 | 1.75±0.10 | 4.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |
| 1210 | 3.00±0.20 | 3.60±0.20 | 8.00±0.30 | 3.50±0.05 | 1.75±0.10 | 4.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |
| 2010 | 2.80±0.20 | 5.50±0.20 | 12.00±0.30 | 5.50±0.05 | 1.75±0.10 | 4.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |
| 2512 | 3.50±0.20 | 6.70±0.20 | 12.00±0.30 | 5.50±0.05 | 1.75±0.10 | 4.00±0.10 | 2.00±0.05 | 4.00±0.10 | 1.50+0.10/-0 |

unit：mm

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## 9.3 Reel Dimensions




unit:mm

| Size | Packaging Q'ty | A | N | C | D | B | G | T |
|---|---|---|---|---|---|---|---|---|
| 0402 | 10kpcs/Reel | 178.0±2.0 | 60.0±0.5 | 13.0±0.5 | 20(Min.) | 2.0±0.5 | 10.0±1.5 | 14.9max. |
| 0603<br>0805<br>1206<br>1210 | 5kpcs/Reel | 178.0±2.0 | 60.0±0.5 | 13.0±0.5 | 20(Min.) | 2.0±0.5 | 10.0±1.5 | 14.9max. |
| | 10kpcs/Reel | 254.0±2.0 | 100.0±1.0 | 13.5±0.5 | 20(Min.) | 2.0±0.5 | 10.0±1.5 | 14.9max. |
| | 20kpcs/Reel | 330.0±2.0 | 100.0±1.0 | 13.5±0.5 | 20(Min.) | 2.0±0.5 | 10.0±1.5 | 14.9max. |
| 2010<br>2512 | 4kpcs/Reel | 178.0±2.0 | 60.0±0.5 | 13.0±0.5 | 20(Min.) | 2.0±0.5 | 13.8±1.5 | 16.7max. |
| | 8kpcs/Reel | 254.0±2.0 | 100.0±0.5 | 13.5±0.5 | 20(Min.) | 2.0±0.5 | 13.8±1.5 | 20.0max. |
| | 16kpcs/Reel | 330.0±2.0 | 100.0±1.0 | 13.5±0.5 | 20(Min.) | 2.0±0.5 | 13.8±1.5 | 20.0max. |

## 10. Storage &. Handling

… Products are recommended to be used up within one year as ensured shelf life.
Check solder ability in case shelf life extension is needed.

… To store products with following condition:
Temperature:5 to 40℃；Humidity: 20 to 70% relative humidity.

*Precaution for use：*

*The standard AEC-Q200 series resistors are mainly used on general automotive equipment without safety considerations. Please select SAFETY concern type or contact our company in advanced if you intend to use resistor for designing the equipment which may damage itself and the safety of third party. If necessary, please consider to add the protect circuit in devising process and obtaining fully safety evaluation. The contents of the acknowledgments only used for our parent company, marketing subsidiaries and official marketing agents who purchase our products. Not applicable for the other nonofficial channels.*

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Appendix

## ■ 0603 1% Marking Table (Table 1)

| Code | E48 | E96 | Code | E48 | E96 | Code | E48 | E96 | Code | E48 | E96 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 01 | 100 | 100 | 25 | 178 | 178 | 49 | 316 | 316 | 73 | 562 | 562 |
| 02 | | 102 | 26 | | 182 | 50 | | 324 | 74 | | 576 |
| 03 | 105 | 105 | 27 | 187 | 187 | 51 | 332 | 332 | 75 | 590 | 590 |
| 04 | | 107 | 28 | | 191 | 52 | | 340 | 76 | | 604 |
| 05 | 110 | 110 | 29 | 196 | 196 | 53 | 348 | 348 | 77 | 619 | 619 |
| 06 | | 113 | 30 | | 200 | 54 | | 357 | 78 | | 634 |
| 07 | 115 | 115 | 31 | 205 | 205 | 55 | 365 | 365 | 79 | 649 | 649 |
| 08 | | 118 | 32 | | 210 | 56 | | 374 | 80 | | 665 |
| 09 | 121 | 121 | 33 | 215 | 215 | 57 | 383 | 383 | 81 | 681 | 681 |
| 10 | | 124 | 34 | | 221 | 58 | | 392 | 82 | | 698 |
| 11 | 127 | 127 | 35 | 226 | 226 | 59 | 402 | 402 | 83 | 715 | 715 |
| 12 | | 130 | 36 | | 232 | 60 | | 412 | 84 | | 732 |
| 13 | 133 | 133 | 37 | 237 | 237 | 61 | 422 | 422 | 85 | 750 | 750 |
| 14 | | 137 | 38 | | 243 | 62 | | 432 | 86 | | 768 |
| 15 | 140 | 140 | 39 | 249 | 249 | 63 | 442 | 442 | 87 | 787 | 787 |
| 16 | | 143 | 40 | | 255 | 64 | | 453 | 88 | | 806 |
| 17 | 147 | 147 | 41 | 261 | 261 | 65 | 464 | 464 | 89 | 825 | 825 |
| 18 | | 150 | 42 | | 267 | 66 | | 475 | 90 | | 845 |
| 19 | 154 | 154 | 43 | 274 | 274 | 67 | 487 | 487 | 91 | 866 | 866 |
| 20 | | 158 | 44 | | 280 | 68 | | 499 | 92 | | 887 |
| 21 | 162 | 162 | 45 | 287 | 287 | 69 | 511 | 511 | 93 | 909 | 909 |
| 22 | | 165 | 46 | | 294 | 70 | | 523 | 94 | | 931 |
| 23 | 169 | 169 | 47 | 301 | 301 | 71 | 536 | 536 | 95 | 953 | 953 |
| 24 | | 174 | 48 | | 309 | 72 | | 549 | 96 | | 976 |

| Code | A | B | C | D | E | F | G | H | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Multiplier | $10^{0}$ | $10^{1}$ | $10^{2}$ | $10^{3}$ | $10^{4}$ | $10^{5}$ | $10^{6}$ | $10^{7}$ | $10^{-1}$ | $10^{-2}$ | $10^{-3}$ |

※ All product specification and data are subject to change without notice.

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